Kyle Matro
CSC 351
Assignment 4

```
3
4    #1
5
6    use hw4;
7    select missions.MissionName, missions.Destination, missions.LaunchYear,
8    missions.outcome, launch_sites.SiteName, organizations.Orgname, rockets.RocketName
9    from missions
10   join rockets
11       on missions.RocketID = rockets.RocketID
12   join organizations
13       on rockets.OrgID = organizations.OrgID
14   join launch_sites
15       on launch_sites.SiteID = missions.SiteID
16   order by LaunchYear desc, MissionName asc;
17
18   #2
19
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

MissionName	Destination	LaunchYear	outcome	SiteName	Orgname	RocketName
Artemis I	Moon	2024	Success	Kennedy Space Center	National Aeronautics and Space Administration	Saturn V
Chang'e 6	Moon	2024	Success	Jiuquan Satellite Launch Center	China National Space Administration	Long March 2F
Crew-8	ISS	2024	Success	Kennedy Space Center	Space Exploration Technologies Corp	Falcon 9
Gaganyaan Test	LEO	2024	Success	Satish Dhawan Space Centre	Indian Space Research Organisation	PSLV
HTV-X1 Test	ISS	2024	Success	Tanegashima Space Center	Japan Aerospace Exploration Agency	H-IIB
Progress MS-27	ISS	2024	Success	Baikonur Cosmodrome	Roscosmos State Corporation	Proton-M
Sentinel-6B	LEO	2024	Success	Vandenberg Space Force Base	Space Exploration Technologies Corp	Falcon 9
StriX-1	LEO	2024	Success	Mahia Launch Complex 1	Rocket Lab	Electron

```
18   #2
19
20   select missions.MissionName, rockets.RocketName, payloads.PayloadName,
21   payloads.PayloadType, payloads.MassKg
22   from payloads
23   join missions
24       on payloads.MissionID = missions.MissionID
25   join rockets
26       on missions.RocketID = rockets.RocketID
27   where missions.LaunchYear = 2024
28   order by payloads.MassKg desc;
29
30   #3
31
32   select launch_sites.SiteName, missions.MissionName, rockets.RocketName,
33   missions.LaunchYear
34   from launch_sites
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MissionName	RocketName	PayloadName	PayloadType	MassKg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

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```
29
30      #3
31
32  •   select launch_sites.SiteName, missions.MissionName, rockets.RocketName,
33      missions.LaunchYear
34      from launch_sites
35      left join missions
36          on launch_sites.SiteID = missions.SiteID
37          and missions.LaunchYear = 2024
38      left join rockets
39          on missions.RocketID = rockets.RocketID
40      order by launch_sites.SiteName asc;
41
42      #4
43
44  •   with org_payload_totals as (
45          select rockets.OrgID, sum(payloads.MassKg) as TotalPayloadMassKg
46          from payloads
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SiteName	MissionName	RocketName	LaunchYear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

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```
#4

with org_payload_totals as (
    select rockets.OrgID, sum(payloads.MassKg) as TotalPayloadMassKg
    from payloads
    join missions
        on payloads.MissionID = missions.MissionID
    join rockets
        on missions.RocketID = rockets.RocketID
    group by rockets.OrgID
)
select organizations.OrgName, organizations.HeadquartersCountry,
organizations.OrgType, org_payload_totals.TotalPayloadMassKg
from organizations
join org_payload_totals
    on organizations.OrgID = org_payload_totals.OrgID
order by org_payload_totals.TotalPayloadMassKg desc;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OrgName	HeadquartersCountry	OrgType	TotalPayloadMassKg
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100

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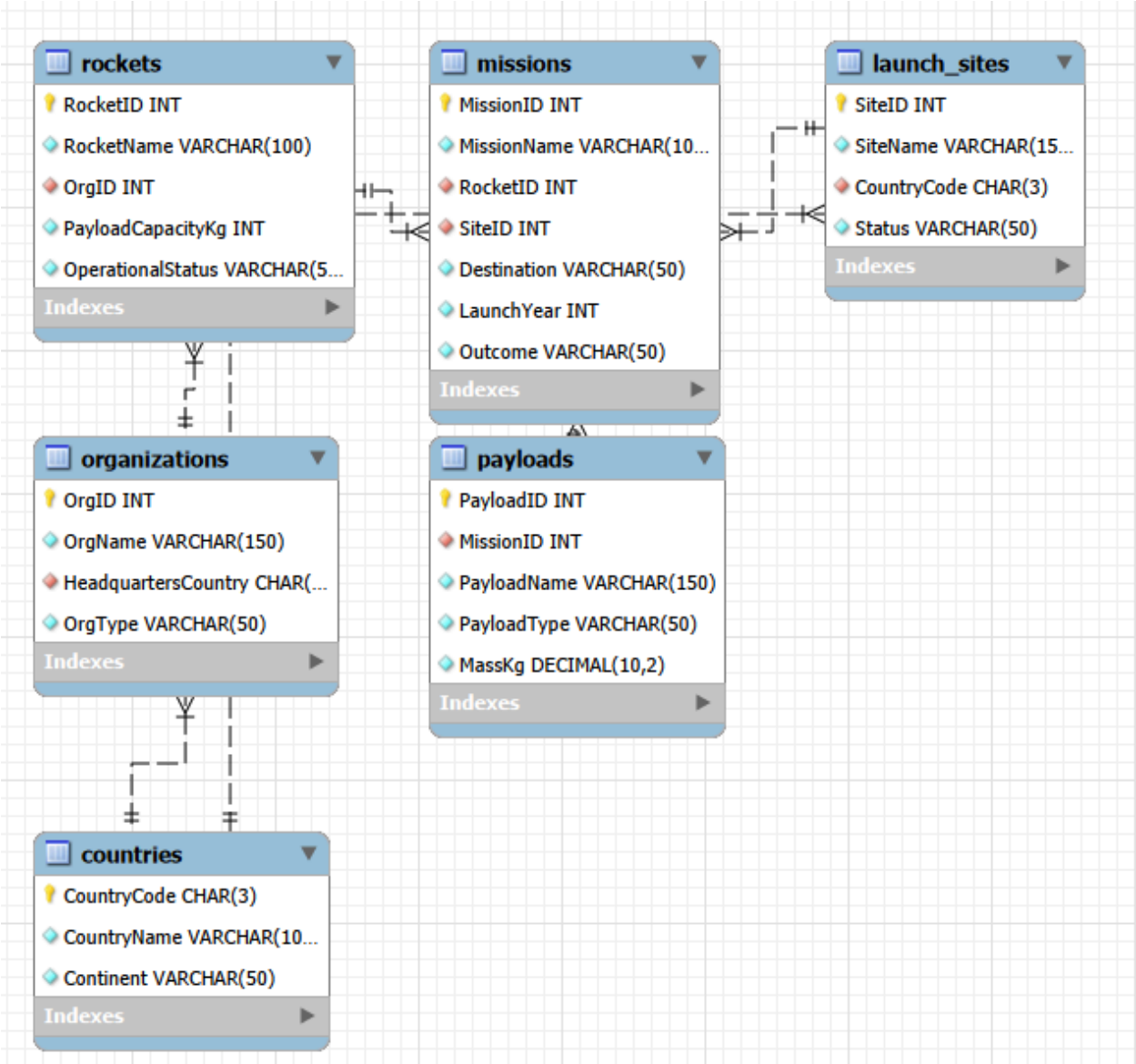
```sql
#5

with country_missions as (
    select launch_sites.CountryCode, count(missions.MissionID) as TotalMissions
    from missions
    join launch_sites
        on missions.SiteID = launch_sites.SiteID
    group by launch_sites.CountryCode
),
country_payloads as (
    select launch_sites.CountryCode, sum(payloads.MassKg) as TotalPayloadMassKg
    from payloads
    join missions
        on payloads.MissionID = missions.MissionID
    join launch_sites
        on missions.SiteID = launch_sites.SiteID
    group by launch_sites.CountryCode
)
select countries.CountryName, country_missions.TotalMissions,
country_payloads.TotalPayloadMassKg
from countries
join country_missions
    on countries.CountryCode = country_missions.CountryCode
join country_payloads
    on countries.CountryCode = country_payloads.CountryCode
order by country_payloads.TotalPayloadMassKg desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CountryName	TotalMissions	TotalPayloadMassKg
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150
New Zealand	1	100

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rockets
RocketID INT
RocketName VARCHAR(100)
OrgID INT
PayloadCapacityKg INT
OperationalStatus VARCHAR(5...
Indexes
missions
MissionID INT
MissionName VARCHAR(10...
RocketID INT
SiteID INT
Destination VARCHAR(50)
LaunchYear INT
Outcome VARCHAR(50)
Indexes
launch_sites
SiteID INT
SiteName VARCHAR(15...
CountryCode CHAR(3)
Status VARCHAR(50)
Indexes
organizations
OrgID INT
OrgName VARCHAR(150)
HeadquartersCountry CHAR(...
OrgType VARCHAR(50)
Indexes
payloads
PayloadID INT
MissionID INT
PayloadName VARCHAR(150)
PayloadType VARCHAR(50)
MassKg DECIMAL(10,2)
Indexes
countries
CountryCode CHAR(3)
CountryName VARCHAR(10...
Continent VARCHAR(50)
Indexes